

Capstone Project Report 1 • Code: AIP491

An Offline Intelligent Vietnamese Input Method Based on Deep Learning

Project

NextKey

GROUP MEMBERS

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PART I

Project Overview & Research Background



Motivation, problem formulation, objectives, and scope of NextKey.

NextKey Overview

Redefining Vietnamese Mobile Typing

Traditional input methods like Telex and VNI require typing explicit tone marks and diacritics. NextKey introduces a novel **continuous Latin input paradigm** (similar to Chinese Pinyin).

- ✓ **Continuous Unaccented Input:** Type continuous text without spaces or tone marks (e.g., "toidihoc" → "tôi đi học").
- ✓ **End-to-End Deep Learning:** Jointly predicts word boundaries and restores diacritics in real-time.
- ✓ **100% Edge-based:** Fully offline execution ensuring ultra-low latency (<50ms) and absolute user privacy.



Project Motivation & Value



1. Language Inclusion

Language Scaffold for All:

- **Children & Students:** Type thoughts freely without getting stuck on complex spelling rules.
- **Elderly Users:** Eliminate the strain of memorizing tone key combinations on small screens.
- **Foreign Learners:** Express Vietnamese naturally without diacritic barrier.



2. Ergonomics & Speed

Maximize Efficiency:

- **20%–40% Keystroke Savings (KS):** Significantly reduces physical typing effort.
- **Continuous Flow:** Eliminates manual spacebar and accent key presses.
- **Health Impact:** Reduces Repetitive Strain Injury (RSI) risks.



3. Seamless Multilingual

Zero "Dinh Dấu"

Errors:

- **No English Corruption:** Prevents unwanted accents in words like "case" → "cáe", "deadline", or "meeting".
- **Zero Layout Switching:** Seamlessly type mixed VI-EN sentences without toggling input modes.

Existing Solutions & Limitations



Telex / VNI

Rule-based deterministic mapping. Requires high cognitive effort, explicit tone keys (s, f, r, x, j), and high keystroke count.



N-Gram & HMM

Statistical models. Suffer from the curse of dimensionality and fail when handling completely unsegmented, continuous input.



RNN / Bi-LSTM

Captures sequential context well, but high inference latency makes real-time keyboard deployment difficult.



Transformers

State-of-the-art accuracy (BARTpho/T5), but massive parameter footprint prevents offline mobile deployment without heavy quantization.

Problem Statement

Dual Linguistic Challenge

Given continuous unaccented Latin text without spaces:

```
Input: "toidihoc"  
Output: "tôi đi học" / "tối đi học"
```

Must resolve **homograph ambiguities** and **boundary segmentation** simultaneously based on global semantic context.

Edge Computing Constraints

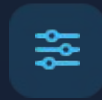
- **Strict Latency:** Must deliver inference under **< 50 ms** per keystroke event.
- **Zero Cloud Dependency:** Must execute completely offline on mobile devices.
- **Resource Bounds:** Optimized for limited mobile RAM and battery consumption.

Key Objectives & Impact



Data & Modeling

Build large-scale parallel synthetic corpus (diacritic stripping, space erasure, noise modeling). Fine-tune Seq2Seq Transformer (T5/ByT5).



Model Compression

Apply Knowledge Distillation, Beam Search optimization, and INT8 Quantization for lightweight mobile edge deployment.



Ergonomic Savings

Achieve estimated **20% - 40%** **Keystroke Savings (KS)**, significantly increasing typing speed.



Absolute Privacy

100% local device execution guarantees sensitive keystrokes and personal messages are never sent over the internet.

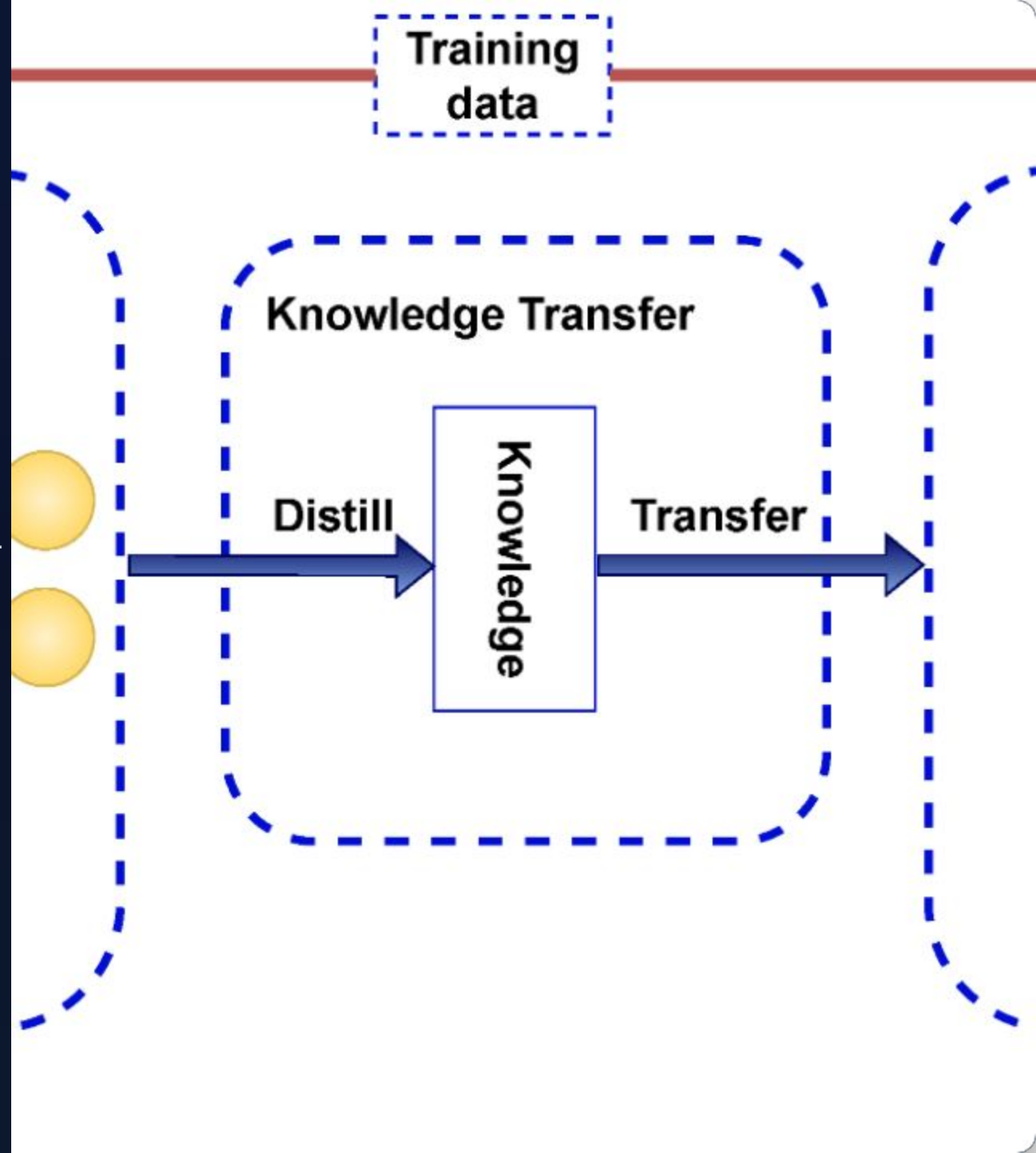
Project Scope

✅ In-Scope Deliverables

- End-to-End Deep Learning Seq2Seq Model architecture.
- Synthetic data augmentation & typo noise pipeline.
- Quantized ONNX/TFLite edge inference backend API.
- Interactive proof-of-concept keyboard interface.

❌ Out-of-Scope

- OS-level low-level driver development for iOS/Android native installation.
- Voice recognition and historical ancient scripts (Chữ Nôm).



PART II

Project Management & Roadmap



Team roles, work breakdown structure, timeline, and risk management strategy.

Team Structure & Roles

Role	Assignee	Core Responsibilities
Leader / AI Researcher	Trần Đình Gia (SE181550)	Project management, sequence modeling formulation, multi-task learning design, hyperparameter tuning & training supervision.
Edge Systems & Optimization	Trương Nhật Quang (DE180544)	Decoding strategy optimization (Beam Search), Knowledge Distillation, INT8 quantization, latency benchmarking on edge hardware.
Data Engineering & Backend	Nguyễn Xuân Cường (DE180528)	Vietnamese corpus processing, noise generation/typo simulation pipeline, offline inference backend API development.

Work Breakdown Structure (20 Weeks)

Phase 1: W1-4

Data & Literature

Corpus collection, synthetic noise pipeline (diacritic/space removal), tokenization setup.

Phase 2: W5-9

Model Development

T5-based Seq2Seq fine-tuning, hyperparameter tuning, error analysis & evaluation.

Phase 3: W10-14

Optimization

Knowledge Distillation, INT8 quantization, BLEU / WER & latency (<50ms) benchmarking.

Phase 4: W15-17

System Backend

Offline inference backend engine, keyboard interaction API, integration tests.

Phase 5: W18-20

Validation & Defense

User testing, Keystroke Savings evaluation, thesis documentation & defense prep.

Risk Management Strategy

Identified Risk	Severity	Prob.	Mitigation Strategy
Domain Generalization Gap Poor accuracy on real noisy typing	High	Med	Include synthetic typo noise (fat-finger errors) and informal vocabulary (teencode) during training.
High Inference Latency Inference > 50ms disrupts typing experience	High	Med	Aggressively apply Knowledge Distillation, INT8 quantization, and evaluate lightweight Transformer variants.
Compute Resource Limits Slow training iterations on large models	Med	High	Leverage cloud GPU allocations (Colab, Kaggle) and FP16 mixed-precision training.

EVALUATION METRICS & QA

Core Metrics & Expected Deliverables

< 50 ms

Inference Latency

20% - 40%

Keystroke Savings (KS)

WER / BLEU

Linguistic Accuracy

100%

Offline

Edge Data Privacy

Thank You for Listening!

Questions & Comments Are Welcome